

Multiple Choice Questions: Percentages

Mathematics Practice Module

December 30, 2025

1 Practice Questions

- Q1.** If x is 80% of y , then what percent of x is y ?
A) 75% B) 80% C) 100% D) 125%
- Q2.** A student has to obtain 33% of the total marks to pass. He got 125 marks and failed by 40 marks. The maximum marks are:
A) 300 B) 500 C) 800 D) 1000
- Q3.** If the price of a commodity is increased by 25%, by how much percent must a consumer reduce the consumption so as not to increase the expenditure?
A) 15% B) 20% C) 25% D) 30%
- Q4.** 0.15% of $33\frac{1}{3}\%$ of \$10,000 is:
A) \$5 B) \$50 C) \$150 D) \$0.05
- Q5.** If A's income is 25% more than B's income, then by how much percent is B's income less than A's income?
A) 15% B) 20% C) 25% D) 30%
- Q6.** The population of a town increased from 1,75,000 to 2,62,500 in a decade. The average percent increase of population per year is:
A) 4.37% B) 5% C) 6% D) 8.75%
- Q7.** In an election between two candidates, one got 55% of the total valid votes. 20% of the votes were invalid. If the total number of votes was 7500, the number of valid votes that the other candidate got, was:
A) 2700 B) 2900 C) 3000 D) 3100
- Q8.** A fruit seller had some apples. He sells 40% apples and still has 420 apples. Originally, he had:
A) 588 apples B) 600 apples C) 672 apples D) 700 apples
- Q9.** What percentage of numbers from 1 to 70 have 1 or 9 in the unit's digit?
A) 1% B) 14% C) 20% D) 21%
- Q10.** If 75% of a number is added to 75, then the result is the number itself. The number is:
A) 50 B) 60 C) 300 D) 400
- Q11.** A number is increased by 20% and then decreased by 20%. The final value of the number:
A) Does not change B) Decreases by 2% C) Decreases by 4% D) Increases by 4%

- Q12.** In a certain school, 20% of students are below 8 years of age. The number of students above 8 years of age is $\frac{2}{3}$ of the number of students of 8 years of age which is 48. What is the total number of students in the school?
 A) 72 B) 80 C) 100 D) 120
- Q13.** Two numbers A and B are such that the sum of 5% of A and 4% of B is two-third of the sum of 6% of A and 8% of B. Find the ratio of A : B.
 A) 2 : 3 B) 1 : 1 C) 3 : 4 D) 4 : 3
- Q14.** A's salary is 40% of B's salary, which is 25% of C's salary. What percentage of C's salary is A's salary?
 A) 5% B) 10% C) 15% D) 20%
- Q15.** If the radius of a circle is diminished by 10%, then its area is diminished by:
 A) 10% B) 19% C) 20% D) 36%
- Q16.** A person spends 40% of his salary on food, 20% on house rent, 10% on entertainment and 10% on conveyance. If his savings at the end of a month are \$1500, then his monthly salary is:
 A) \$6000 B) \$7500 C) \$8000 D) \$10000
- Q17.** When a number is increased by 24, it becomes 115% of itself. What is the number?
 A) 160 B) 200 C) 250 D) 300
- Q18.** 5% of income of A is equal to 15% of income of B and 10% of income of B is equal to 20% of income of C. If C's income is \$2000, then the total income of A, B and C is:
 A) \$6000 B) \$14000 C) \$18000 D) \$20000
- Q19.** In an examination, 35% of the total students failed in Hindi, 45% failed in English and 20% failed in both. The percentage of those who passed in both the subjects is:
 A) 10% B) 20% C) 30% D) 40%
- Q20.** The value of a machine depreciates at the rate of 10% every year. It was purchased 3 years ago. If its present value is \$8748, its purchase price was:
 A) \$10000 B) \$12000 C) \$14000 D) \$16000
- Q21.** If the numerator of a fraction is increased by 200% and the denominator is increased by 350%, the resultant fraction is $\frac{5}{12}$. What was the original fraction?
 A) $\frac{5}{8}$ B) $\frac{5}{9}$ C) $\frac{7}{12}$ D) $\frac{11}{12}$
- Q22.** A batsman scored 110 runs which included 3 boundaries and 8 sixes. What percent of his total score did he make by running between the wickets?
 A) 45% B) $45\frac{5}{11}\%$ C) 50% D) $54\frac{6}{11}\%$
- Q23.** 40% of a number is equal to $\frac{3}{5}$ of another number. What is the ratio between the first number and the second number?
 A) 2 : 3 B) 3 : 2 C) 7 : 9 D) 3 : 4
- Q24.** The price of sugar having gone down by 10%, a consumer can buy 5 kg more sugar for \$270. The difference between the original and reduced price per kg is:
 A) \$0.50 B) \$0.60 C) \$0.75 D) \$1.00

- Q25.** Out of 450 students of a school, 325 play football, 175 play cricket and 50 neither play football nor cricket. How many students play both?
 A) 50 B) 75 C) 100 D) 125
- Q26.** If 20% of $a = b$, then $b\%$ of 20 is the same as:
 A) 4% of a B) 5% of a C) 20% of a D) None of these
- Q27.** Three candidates contested an election and received 1136, 7636 and 11628 votes respectively. What percentage of the total votes did the winning candidate get?
 A) 57% B) 60% C) 65% D) 90%
- Q28.** Gauri went to the stationers and bought things worth \$25, out of which 30 cents went on sales tax on taxable purchases. If the tax rate was 6%, then what was the cost of the tax-free items?
 A) \$15 B) \$15.70 C) \$19.70 D) \$20
- Q29.** A man spends 35% of his income on food, 25% on children's education and 80% of the remaining on house rent. What percent of his income is he left with?
 A) 6% B) 8% C) 10% D) 12%
- Q30.** If A's height is 10% more than B's height, by how much percent is B's height less than A's height?
 A) 10% B) $10\frac{1}{11}\%$ C) $9\frac{1}{11}\%$ D) 9%

2 Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	7	A	13	D	19	D	25	C
2	B	8	D	14	B	20	B	26	A
3	B	9	C	15	B	21	A	27	A
4	A	10	C	16	B	22	B	28	C
5	B	11	C	17	A	23	B	29	B
6	B	12	C	18	C	24	B	30	C

3 Brief Explanatory Hints

- **Q2:** Pass marks = $125 + 40 = 165$. Since $33\% = 165$, then $100\% = (165/33) \times 100 = 500$.
- **Q3:** Formula: $[\frac{r}{100+r} \times 100]\%$. Here $\frac{25}{125} \times 100 = 20\%$.
- **Q11:** Result of $x\%$ increase and $x\%$ decrease is always a decrease of $\frac{x^2}{100}\%$. Here $\frac{20^2}{100} = 4\%$.
- **Q19:** Using Set Theory: Total fail = $(35 + 45 - 20) = 60\%$. Pass % = $100 - 60 = 40\%$.
- **Q28:** Tax is \$0.30. Since rate is 6%, taxable purchase = $(0.30/0.06) = \$5$. Total spent = \$25. Tax-free = $25 - 5 - 0.30 = \$19.70$.